Lab assignment 3

(Trajectory estimation using IMU)

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Due on: 18-12-2026, 23:59:59
To be addressed in groups of up to 3 students

Syllabus

Estimating a trajectory corresponding to the movement of an object/robot is a task relevant in multiple domains, e.g., aerospace, logistics, healthcare. The estimates are obtained after data obtained from sensors rigidly attached to the robot.

In this lab, students will perform arbitrary movements holding a Grove IMU sensor and will (try to) estimate the corresponding trajectory from the data acquired. Additionally, another objective is to work on techniques to improve the quality of the data from the sensors.

The Grove IMU (fig.1) has 9 dof (3D accelerometer + 3D rate-gyro + 3D magnetometer). For the purpose of the lab we will consider only the accelerometer and the rate-gyro.

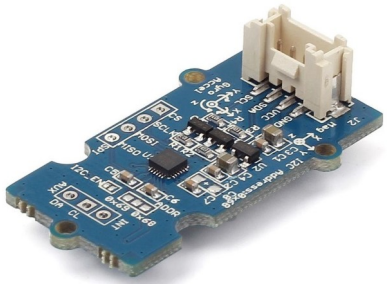


Fig. 1 - Grove 9-dof IMU.

The IMU is attached to an Arduino Uno, that handles the data acquisition and sends it, via serial RS232 protocol, to a remote (laptop) computer – the computational power of Arduino Uno is limited. It is thus required that students have (i) one free USB port, where to connect the Arduino, and (ii) the Arduino IDE (freely available online) installed in their laptops to run the data acquisition software. The software to run on the Arduino is available from the course webpage and, if needed, can be modified for the purpose of the lab. Similarly, a Python script is available to collect the data from the Arduino via a serial port.

Tasks

The tasks below must be answered independently in the report of the lab. There is no page limit but 6 A4 pages should be enough. Also, the report should avoid verbose text.

1. Indicate adequate mathematical models for the accelerometer and for the rate-gyro. Clearly indicate the parameters to be estimated.

2. Perform experiments to estimate such parameters. Don't forget to record videos of the experiments. Comment the results.

Hint: consider filtering the raw data.

3. Implement an inclinometer using only accelerometer data.

4. Estimate the orientation of the device using only rate-gyro data.

5. Fuse the estimates obtained in 3 and 4 using, for example, Covariance Intersection (CI). Plot the estimates for different values of the CI parameter.

6. Using the best parameter obtained in 5, and holding the device steadily in your hand, perform (i) a waiving gesture, and (ii) a punching gesture. In both cases move the arm/hand slowly. As in the other tasks, the results should be clearly presented in the report.